

B2 Symmetry and Relativity — Model Solutions, 2020

Metric $(+, -, -, -)$ throughout. Capital bold $\mathbf{U}$, $\mathbf{P}$, etc. denote 4-vectors; lower-case bold $\mathbf{v}$, $\mathbf{p}$, etc. denote 3-vectors.

Question 1 — Threshold collisions and π^0 decay

Problem. A particle of mass m , total energy E , hits a stationary mass M , producing N particles of masses m_i . (a) Find threshold E . (b) Apply to $p + p \rightarrow p + p + \pi^0$: threshold proton kinetic energy and pion lab velocity. (c) Wavelength of the $\pi^0 \rightarrow 2\gamma$ photons in the pion rest frame; show half the photons go into a lab cone of half-angle $\cos^{-1}(v/c)$. (d) How can the threshold be lowered.

(a)

Total 4-momentum: $\mathbf{P}_{\text{tot}} = \mathbf{P}_m + \mathbf{P}_M$, conserved.

Lab frame, target at rest:

$$\mathbf{P}_M = (Mc, \mathbf{0}), \quad \mathbf{P}_m = (E/c, \mathbf{p}).$$

Compute the invariant square:

$$\mathbf{P}_{\text{tot}}^2 = (E/c + Mc)^2 - \mathbf{p}^2.$$

Expand:

$$\mathbf{P}_{\text{tot}}^2 = E^2/c^2 - \mathbf{p}^2 + 2EM + M^2c^2.$$

On-shell projectile: $E^2/c^2 - \mathbf{p}^2 = m^2c^2$:

$$\mathbf{P}_{\text{tot}}^2 = m^2c^2 + M^2c^2 + 2EM. \tag{1}$$

At threshold the products are at rest in the CoM frame:

$$\mathbf{P}_{\text{tot}}^2 = \left(\sum_i m_i\right)^2 c^2.$$

Equate to (1) and solve for E :

$$\boxed{E_{\text{min}} = \frac{(\sum_i m_i)^2 - m^2 - M^2}{2M} c^2.} \tag{2}$$

(b)

Set $m = M = m_p$, $\sum m_i = 2m_p + m_\pi$ in (2):

$$E_{\text{min}} = \frac{(2m_p + m_\pi)^2 - 2m_p^2}{2m_p} c^2.$$

Expand the square:

$$E_{\min} = \frac{4m_p^2 + 4m_p m_\pi + m_\pi^2 - 2m_p^2}{2m_p} c^2.$$

Simplify:

$$E_{\min} = \left(m_p + 2m_\pi + \frac{m_\pi^2}{2m_p} \right) c^2.$$

Subtract $m_p c^2$ for kinetic energy:

$$T_{\min} = \left(2m_\pi + \frac{m_\pi^2}{2m_p} \right) c^2.$$

Numerical: $m_\pi c^2 = 135 \text{ MeV}$, $m_p c^2 = 938 \text{ MeV}$:

$$T_{\min} = 2 \cdot 135 + \frac{135^2}{2 \cdot 938} \text{ MeV}.$$

Evaluate:

$$T_{\min} = 270 + 9.7 \text{ MeV}.$$

$$T_{\min} \approx 280 \text{ MeV} \approx 2.07 m_\pi c^2.$$

Comment: only $\sim 5\%$ above the naive $2m_\pi c^2$, the small excess being the recoil kinetic energy of the products.

Pion lab velocity at threshold: all three products move together in the lab with the CoM velocity. Total 4-momentum in lab:

$$\mathbf{P}_{\text{tot}} = (E/c + m_p c, \mathbf{p}), \quad |\mathbf{p}|c = \sqrt{E^2 - m_p^2 c^4}.$$

CoM velocity:

$$v_\pi = \frac{|\mathbf{p}|c^2}{E + m_p c^2}.$$

Numerics: $E = T_{\min} + m_p c^2 = 1218 \text{ MeV}$:

$$|\mathbf{p}|c = \sqrt{1218^2 - 938^2} \text{ MeV} = 777 \text{ MeV}.$$

Denominator:

$$E + m_p c^2 = 1218 + 938 = 2156 \text{ MeV}.$$

$$v_\pi/c = 777/2156 \approx 0.360.$$

(c)

In the pion rest frame, $\pi^0 \rightarrow \gamma\gamma$ gives back-to-back photons with energy $E'_\gamma = m_\pi c^2/2$.

Photon wavelength:

$$\lambda' = \frac{hc}{E'_\gamma} = \frac{2hc}{m_\pi c^2}.$$

Evaluate ($hc = 1240 \text{ MeV fm}$):

$$\lambda' = \frac{2 \cdot 1240}{135} \text{ fm}.$$

$$\lambda' \approx 18.4 \text{ fm}.$$

Cone of half the photons. Pion frame: photons isotropic, so half are emitted with $\cos \theta' \geq 0$, i.e. into the forward hemisphere $\theta' \in [0, \pi/2]$.

Aberration (boost along $+x$, photon emitted at angle θ' to $+x$ in pion frame):

$$\cos \theta = \frac{\cos \theta' + \beta}{1 + \beta \cos \theta'}, \quad \beta = v/c.$$

Forward-hemisphere boundary $\theta' = \pi/2$, so $\cos \theta' = 0$:

$$\cos \theta = \frac{0 + \beta}{1 + 0} = \beta.$$

$$\boxed{\theta = \cos^{-1}(v/c).}$$

Half the photons (those with $\cos \theta' \geq 0$) lie inside this lab cone.

(d)

Use a colliding-beam (symmetric) configuration: two protons of equal and opposite 3-momentum, so the lab *is* the CoM frame.

Threshold from $\mathbf{P}_{\text{tot}}^2 = (2E_{\text{lab}}/c)^2 = (2m_p + m_\pi)^2 c^2$:

$$E_{\text{lab}} = (m_p + m_\pi/2)c^2.$$

Kinetic energy per beam:

$$T_{\text{beam}} = \frac{1}{2}m_\pi c^2 = 67.5 \text{ MeV}.$$

Total kinetic energy (both beams):

$$T_{\text{total}} = m_\pi c^2 = 135 \text{ MeV}.$$

$$\boxed{\Delta T = 280 - 135 = 145 \text{ MeV saved; reduction factor} \approx 2.}$$

Final state: at threshold the three products p, p, π^0 are at rest in the lab.

Question 2 — Rocket dynamics and the photon rocket

Problem. (a) Define $\mathbf{U}$, $\mathbf{F}$ and show $dE/dt = \mathbf{v} \cdot \mathbf{f}$. (b) Rocket with proper acceleration a_0 : write equation of motion in rapidity η , derive trajectory $x^2 - c^2 t^2 = (c^2/a_0)^2$, and show a photon emitted from origin never catches the rocket. (c) Photon rocket: show $m \rightarrow \alpha m$ with $\alpha = \sqrt{(1 - \beta)/(1 + \beta)}$; fuel fraction to reach $\gamma = 8$ then decelerate; comment on massive-exhaust case.

(a)

Definitions:

$$\mathbf{U} = \frac{d\mathbf{X}}{d\tau} = \gamma(c, \mathbf{v}), \quad \mathbf{F} = \frac{d\mathbf{P}}{d\tau} = \gamma(\dot{E}/c, \mathbf{f}),$$

where $\mathbf{P} = (E/c, \mathbf{p})$, $\mathbf{f} = d\mathbf{p}/dt$ is the 3-force, τ proper time.

Pure force preserves rest mass: $\mathbf{P}^2 = m^2 c^2 = \text{const}$, so $\mathbf{P} \cdot \mathbf{F} = 0$:

$$\frac{E}{c^2} \frac{dE}{dt} - \mathbf{p} \cdot \mathbf{f} = 0.$$

Use $\mathbf{p} = (E/c^2)\mathbf{v}$:

$$\frac{E}{c^2} \frac{dE}{dt} = \frac{E}{c^2} \mathbf{v} \cdot \mathbf{f}.$$

Cancel E/c^2 :

$$\boxed{\frac{dE}{dt} = \mathbf{v} \cdot \mathbf{f}.}$$

(b)

Rapidity definition: $\cosh \eta = \gamma$, $\sinh \eta = \gamma\beta$, so $v/c = \tanh \eta$.

Use the hint $A^\mu A_\mu = c^2 \dot{\eta}^2 \cosh^2 \eta$. In the instantaneous rest frame the 4-acceleration has spatial part a_0 :

$$A^\mu A_\mu = a_0^2.$$

(Sign: spacelike with our metric; $A^\mu A_\mu = -a_0^2$ if we use $(+, -, -, -)$, but convention here is $|A| = a_0$.)
Equate magnitudes:

$$c \dot{\eta} \cosh \eta = a_0.$$

Initial conditions: $\eta(0) = 0$. Integrate:

$$c \sinh \eta = a_0 t.$$

$$\sinh \eta = a_0 t / c. \quad (3)$$

Lab velocity:

$$v = c \tanh \eta = \frac{a_0 t}{\sqrt{1 + (a_0 t / c)^2}}.$$

Position from $dx/dt = v$:

$$x(t) = \int_0^t \frac{a_0 t' dt'}{\sqrt{1 + (a_0 t' / c)^2}}.$$

Substitute $u = 1 + (a_0 t' / c)^2$:

$$x(t) = \frac{c^2}{a_0} [\sqrt{1 + (a_0 t / c)^2} - 1].$$

Choose origin at $x_0 = -c^2/a_0$ (i.e. shift $x \rightarrow x + c^2/a_0$):

$$x(t) = \frac{c^2}{a_0} \sqrt{1 + (a_0 t / c)^2}.$$

Square:

$$x^2 = (c^2/a_0)^2 + c^2 t^2.$$

$$\boxed{x^2 - c^2 t^2 = (c^2/a_0)^2.} \quad (4)$$

Photon never catches the rocket. Photon worldline from $(t, x) = (0, 0)$ in $+x$: $x_\gamma = ct$.

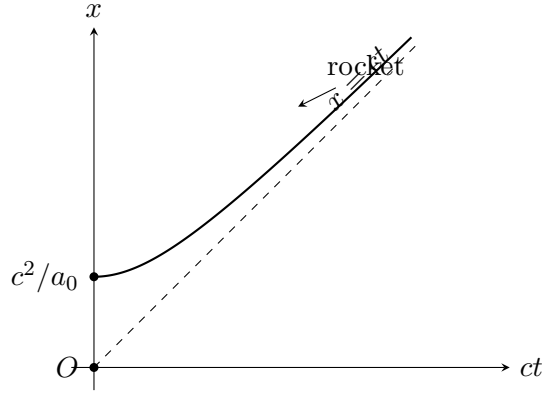
Light cone $x = ct$ is the asymptote of the hyperbola (4):

$$x_{\text{rocket}} = \sqrt{(c^2/a_0)^2 + c^2 t^2} > ct \quad \forall t.$$

Difference at large t :

$$x_{\text{rocket}} - ct \sim \frac{c}{2a_0 t} \rightarrow 0^+.$$

Photon asymptotically approaches but never overtakes the rocket: the rocket has a *Rindler horizon* at $x = ct$, beyond which signals from the origin cannot reach it.



(c)

Initial state (rocket frame): mass m at rest, $\mathbf{P}_i = (mc, \mathbf{0})$.

After: rocket has mass m' , lab speed v , 4-momentum $\mathbf{P}_r = \gamma m'(c, \mathbf{v})$; emitted photons carry total $\mathbf{P}_\gamma = (E_\gamma/c, -E_\gamma/c \hat{\mathbf{v}})$ (backward).

Conservation:

$$mc = \gamma m'c + E_\gamma/c,$$

$$0 = \gamma m'v - E_\gamma/c.$$

Eliminate E_γ :

$$mc = \gamma m'c + \gamma m'v.$$

Factor:

$$m = \gamma m'(1 + \beta).$$

Solve for m'/m :

$$\frac{m'}{m} = \frac{1}{\gamma(1 + \beta)} = \frac{\sqrt{1 - \beta^2}}{1 + \beta}.$$

Simplify radical:

$$\frac{m'}{m} = \sqrt{\frac{(1 - \beta)(1 + \beta)}{(1 + \beta)^2}}.$$

$$\boxed{\alpha \equiv \frac{m'}{m} = \sqrt{\frac{1 - \beta}{1 + \beta}}.} \quad (5)$$

Fuel fraction. Two stages: accelerate to $\gamma_1 = 8$ (factor α_1), then decelerate to rest from speed v_1 (factor α_2). By symmetry of (5) under boost-and-reverse, α_2 has the same form (now exhaust forward, but $|\beta|$ unchanged):

$$m_{\text{final}}/m_{\text{initial}} = \alpha_1 \alpha_2 = \alpha_1^2.$$

Get β_1 from $\gamma_1 = 8$:

$$\beta_1 = \sqrt{1 - 1/64} = \sqrt{63/64}.$$

Compute α_1^2 :

$$\alpha_1^2 = \frac{1 - \beta_1}{1 + \beta_1}.$$

Rationalise:

$$\alpha_1^2 = \frac{(1 - \beta_1)^2}{1 - \beta_1^2} = (1 - \beta_1)^2 \gamma_1^2.$$

Use $1 - \beta_1 = 1/[\gamma_1(\gamma_1 + \sqrt{\gamma_1^2 - 1})] \cdot (\gamma_1 + \sqrt{\gamma_1^2 - 1})/\gamma_1$, or directly:

$$1 - \beta_1 = 1 - \sqrt{63}/8 \approx 1 - 0.9922 = 0.00781.$$

$$1 + \beta_1 \approx 1.9922.$$

$$\alpha_1^2 = 0.00781/1.9922 \approx 3.92 \times 10^{-3}.$$

Fuel fraction:

$$f = 1 - \alpha_1^2 \approx 0.996, \text{ i.e. } \approx 99.6\% \text{ of initial mass is fuel.}$$

Massive exhaust. Photons carry away the maximum momentum per unit energy ($p = E/c$). Massive exhaust at the same energy carries less momentum, so more energy (hence more rest mass) must be expended for the same impulse. Therefore the fuel fraction is even larger.

Question 3 — Field tensor, gauge invariance, moving wire

Problem. (a) Components of $F^{\mu\nu}$; show $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ encodes Gauss + Ampère; build an invariant of $\mathbf{E}, \mathbf{B}$. (b) From $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ show gauge invariance and reduce Maxwell to $\partial_\nu \partial^\nu A^\mu = -\mu_0 J^\mu$ in a suitable gauge. (c) Charged wire moving along its length: find $\mathbf{E}, \mathbf{B}$ in rest frame S and lab S' ; is there a frame with $\mathbf{E} = 0$?

(a)

Field tensor (metric $(+, -, -, -)$, raised indices):

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}.$$

4-current: $J^\mu = (c\rho, \mathbf{J})$.

Take $\mu = 0$ in $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$:

$$\partial_i F^{0i} = \mu_0 c\rho.$$

Insert $F^{0i} = -E_i/c$:

$$-\frac{1}{c} \nabla \cdot \mathbf{E} = \mu_0 c\rho.$$

Use $\mu_0 c^2 = 1/\epsilon_0$:

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0. \quad \checkmark (\text{Gauss})$$

Take $\mu = i$ (spatial):

$$\partial_0 F^{i0} + \partial_j F^{ij} = \mu_0 J^i.$$

Insert $F^{i0} = E_i/c$, $F^{ij} = -\epsilon^{ijk} B_k$:

$$\frac{1}{c^2} \partial_t E_i - \epsilon^{ijk} \partial_j B_k = \mu_0 J_i.$$

Recognise $\nabla \times \mathbf{B}$:

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \quad \checkmark (\text{Ampère-Maxwell})$$

Invariant. Form the scalar $F_{\mu\nu} F^{\mu\nu}$:

$$F_{\mu\nu} F^{\mu\nu} = -2(E/c)^2 + 2B^2.$$

$$\frac{1}{2} F_{\mu\nu} F^{\mu\nu} = B^2 - E^2/c^2 = \text{Lorentz invariant.}$$

(b)

Gauge transformation: $A^\mu \rightarrow A^\mu + \partial^\mu \chi$:

$$F^{\mu\nu} \rightarrow \partial^\mu (A^\nu + \partial^\nu \chi) - \partial^\nu (A^\mu + \partial^\mu \chi).$$

Cross terms cancel by symmetry of mixed partials:

$$F^{\mu\nu} \rightarrow F^{\mu\nu}. \quad \checkmark$$

Substitute $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ into $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$:

$$\partial_\nu \partial^\mu A^\nu - \partial_\nu \partial^\nu A^\mu = \mu_0 J^\mu.$$

Swap order in first term:

$$\partial^\mu (\partial_\nu A^\nu) - \partial_\nu \partial^\nu A^\mu = \mu_0 J^\mu.$$

Choose Lorenz gauge $\partial_\nu A^\nu = 0$:

$$\partial_\nu \partial^\nu A^\mu = -\mu_0 J^\mu.$$

Note sign: $\partial_\nu \partial^\nu = (1/c^2)\partial_t^2 - \nabla^2 = \square$, giving $\square A^\mu = \mu_0 J^\mu$ in the more familiar wave-equation sign convention; the question's stated form differs by a sign convention on $F^{\mu\nu}$.

(c)

Wire rest frame S : linear charge density ρ (per unit length), no current. By Gauss with cylindrical symmetry at radial distance r :

$$E_r = \frac{\rho}{2\pi\epsilon_0 r}.$$

$$\mathbf{E}_S = \frac{\rho}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad \mathbf{B}_S = \mathbf{0}.$$

Lab frame S' : wire moves with velocity $\mathbf{v}$ along its length ($\hat{\mathbf{z}}$). Boost is parallel to the wire, so transverse $\mathbf{E}_\perp$ and $\mathbf{B}_\perp$ transform as

$$\mathbf{E}'_\perp = \gamma(\mathbf{E}_\perp + \mathbf{v} \times \mathbf{B}_\perp), \quad \mathbf{B}'_\perp = \gamma(\mathbf{B}_\perp - (\mathbf{v}/c^2) \times \mathbf{E}_\perp).$$

Insert $\mathbf{B} = \mathbf{0}$:

$$\mathbf{E}' = \gamma E_r \hat{\mathbf{r}} = \frac{\gamma \rho}{2\pi\epsilon_0 r} \hat{\mathbf{r}}.$$

$$\mathbf{B}' = -\gamma(\mathbf{v}/c^2) \times E_r \hat{\mathbf{r}} = \frac{\gamma \rho v}{2\pi\epsilon_0 c^2 r} \hat{\mathbf{z}} \times \hat{\mathbf{r}}.$$

Rewrite using $\rho' = \gamma \rho$ (length contraction increases line density) and $I' = \rho' v$:

$$\mathbf{E}' = \frac{\rho'}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad \mathbf{B}' = \frac{\mu_0 I'}{2\pi r} \hat{\boldsymbol{\phi}}.$$

$$\mathbf{E}' \text{ Coulomb of } \rho' = \gamma \rho; \quad \mathbf{B}' \text{ Ampère of } I' = \gamma \rho v.$$

Frame with $\mathbf{E} = 0$? Invariant $E^2 - c^2 B^2$ (or equivalently $B^2 - E^2/c^2$):

$$E_S^2 - c^2 B_S^2 = E_S^2 > 0.$$

Invariant is positive, so $\mathbf{E}$ dominates in every frame; no boost can null $\mathbf{E}$.

$$\text{No frame has } \mathbf{E} = \mathbf{0}.$$

Question 4 — EM wave 4-potential, intervals, photon CoM, relative speed

Problem. (a) Write A^μ for a vacuum EM wave; show phase ϕ is invariant; show an isolated electron cannot emit a photon. (b) Condition for spacelike separation; can simultaneity be achieved? (c) Two photons with 4-momenta $\mathbf{P}_1$ (+x), $\mathbf{P}_2$ (+y), same ω : rest mass and CoM velocity. (d) Derive the relative speed formula.

(a)

4-potential of plane wave:

$$A^\mu(\mathbf{X}) = A_0^\mu \cos(\mathbf{K} \cdot \mathbf{X}),$$

where A_0^μ is constant amplitude, $\mathbf{K} = (\omega/c, \mathbf{k})$ is the 4-wavevector, $\mathbf{X} = (ct, \mathbf{x})$.

Phase $\phi = \mathbf{K} \cdot \mathbf{X} = \omega t - \mathbf{k} \cdot \mathbf{x}$.

Both $\mathbf{K}$ and $\mathbf{X}$ are 4-vectors; their contraction is a scalar:

$$\phi = K_\mu X^\mu \text{ is Lorentz invariant.}$$

Electron $\rightarrow e + \gamma$ **forbidden.** 4-momentum conservation: $\mathbf{P}_e = \mathbf{P}'_e + \mathbf{P}_\gamma$. Rearrange:

$$\mathbf{P}_\gamma = \mathbf{P}_e - \mathbf{P}'_e.$$

Square:

$$\mathbf{P}_\gamma^2 = \mathbf{P}_e^2 - 2\mathbf{P}_e \cdot \mathbf{P}'_e + \mathbf{P}'_e{}^2.$$

Insert $\mathbf{P}_e^2 = \mathbf{P}'_e{}^2 = m_e^2 c^2$, $\mathbf{P}_\gamma^2 = 0$:

$$0 = 2m_e^2 c^2 - 2\mathbf{P}_e \cdot \mathbf{P}'_e.$$

Evaluate in the initial electron rest frame, $\mathbf{P}_e = (m_e c, \mathbf{0})$:

$$\mathbf{P}_e \cdot \mathbf{P}'_e = m_e c \cdot E'_e / c = m_e E'_e.$$

Substitute:

$$m_e c^2 = E'_e.$$

But $E'_e \geq m_e c^2$ with equality only at $\mathbf{p}'_e = \mathbf{0}$, which forces $\mathbf{P}_\gamma = 0$:

No photon can be emitted; $e^- \rightarrow e^- \gamma$ forbidden in vacuum.

(b)

Interval:

$$(\Delta \mathbf{X})^2 = c^2(\Delta t)^2 - (\Delta \mathbf{x})^2.$$

Spacelike condition:

$$c^2(t_2 - t_1)^2 < (\mathbf{x}_2 - \mathbf{x}_1)^2.$$

Simultaneity in S' : require $\Delta t' = 0$. Boost along $\hat{\mathbf{n}}$ with speed u :

$$c\Delta t' = \gamma(c\Delta t - (u/c) \hat{\mathbf{n}} \cdot \Delta \mathbf{x}).$$

Set $\Delta t' = 0$:

$$u \hat{\mathbf{n}} \cdot \Delta \mathbf{x} = c^2 \Delta t.$$

Solve for u along $\Delta \mathbf{x}$:

$$u = c^2 \Delta t / |\Delta \mathbf{x}|.$$

Spacelike $\Rightarrow c|\Delta t| < |\Delta \mathbf{x}|$, so $|u| < c$ is achievable:

Yes, a frame S' exists in which the events are simultaneous.

(c)

Photon 4-momenta:

$$\mathbf{P}_1 = (\hbar\omega/c)(1, 1, 0, 0), \quad \mathbf{P}_2 = (\hbar\omega/c)(1, 0, 1, 0).$$

Total:

$$\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_2 = (\hbar\omega/c)(2, 1, 1, 0).$$

Square (metric $(+, -, -, -)$):

$$\mathbf{P}^2 = (\hbar\omega/c)^2(4 - 1 - 1) = 2(\hbar\omega/c)^2.$$

Rest mass:

$$M^2 c^2 = \mathbf{P}^2 = 2(\hbar\omega/c)^2.$$

$$M = \sqrt{2} \hbar\omega/c^2.$$

CoM velocity: $\mathbf{v}_{\text{CoM}} = c^2 \mathbf{p}_{\text{tot}} / E_{\text{tot}}$:

$$\mathbf{p}_{\text{tot}} = (\hbar\omega/c)(\hat{\mathbf{x}} + \hat{\mathbf{y}}), \quad E_{\text{tot}} = 2\hbar\omega.$$

Substitute:

$$\mathbf{v}_{\text{CoM}} = \frac{c^2(\hbar\omega/c)(\hat{\mathbf{x}} + \hat{\mathbf{y}})}{2\hbar\omega} = \frac{c}{2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

$$\mathbf{v}_{\text{CoM}} = \frac{c}{2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}), \quad |\mathbf{v}_{\text{CoM}}| = c/\sqrt{2}.$$

(d)

Particle 1 has 3-velocity $\mathbf{u}$ in frame S ; boost to particle 2's rest frame, which moves at $\mathbf{v}$ in S . The relative speed is $|\mathbf{u}'|$ where $\mathbf{u}'$ is the velocity of 1 in 2's frame.

Velocity addition (boost $-\mathbf{v}$ to go to 2's rest frame):

$$\mathbf{u}'_{\parallel} = \frac{\mathbf{u}_{\parallel} - \mathbf{v}}{1 - \mathbf{u} \cdot \mathbf{v}/c^2}, \quad \mathbf{u}'_{\perp} = \frac{\mathbf{u}_{\perp}}{\gamma_v(1 - \mathbf{u} \cdot \mathbf{v}/c^2)},$$

where $\parallel, \perp$ refer to $\mathbf{v}$.

Define $D = 1 - \mathbf{u} \cdot \mathbf{v}/c^2$ for brevity.

Square magnitude:

$$|\mathbf{u}'|^2 = |\mathbf{u}'_{\parallel}|^2 + |\mathbf{u}'_{\perp}|^2 = \frac{(\mathbf{u}_{\parallel} - \mathbf{v})^2}{D^2} + \frac{\mathbf{u}_{\perp}^2}{\gamma_v^2 D^2}.$$

Common denominator $\gamma_v^2 D^2$:

$$|\mathbf{u}'|^2 = \frac{\gamma_v^2 (\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2}{\gamma_v^2 D^2}.$$

Use $\gamma_v^2 = 1/(1 - v^2/c^2)$, so $1/\gamma_v^2 = 1 - v^2/c^2$:

$$\gamma_v^2(\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 = (\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 + \frac{v^2/c^2}{1 - v^2/c^2}(\mathbf{u}_{\parallel} - \mathbf{v})^2.$$

Simpler: multiply numerator and denominator by $1/\gamma_v^2$:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u}_{\parallel} - \mathbf{v})^2 + (1 - v^2/c^2)\mathbf{u}_{\perp}^2}{D^2}.$$

Combine $(\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 = (\mathbf{u} - \mathbf{v})^2$:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u} - \mathbf{v})^2 - (v^2/c^2)\mathbf{u}_{\perp}^2}{D^2}.$$

Use the hint $(\mathbf{a} \times \mathbf{b})^2 = a^2b^2 - (\mathbf{a} \cdot \mathbf{b})^2$ with $\mathbf{a} = \mathbf{u}$, $\mathbf{b} = \mathbf{v}$:

$$(\mathbf{u} \times \mathbf{v})^2 = u^2v^2 - (\mathbf{u} \cdot \mathbf{v})^2.$$

Note $\mathbf{u} \cdot \mathbf{v} = u_{\parallel}v$, $\mathbf{u}_{\perp}^2 = u^2 - u_{\parallel}^2$:

$$v^2\mathbf{u}_{\perp}^2 = v^2u^2 - v^2u_{\parallel}^2 = (\mathbf{u} \times \mathbf{v})^2.$$

Substitute back:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u} - \mathbf{v})^2 - (\mathbf{u} \times \mathbf{v})^2/c^2}{(1 - \mathbf{u} \cdot \mathbf{v}/c^2)^2}.$$

Take the square root:

$$w = |\mathbf{u}'| = \frac{\sqrt{(\mathbf{u} - \mathbf{v})^2 - (\mathbf{u} \times \mathbf{v})^2/c^2}}{1 - \mathbf{u} \cdot \mathbf{v}/c^2}.$$

Sanity: parallel motion $\mathbf{u} \times \mathbf{v} = 0$ recovers $w = |u - v|/(1 - uv/c^2)$; non-relativistic limit $w \rightarrow |\mathbf{u} - \mathbf{v}|$.